

Combinatorics Notes

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2022-03-03

Exercise 6d

Throughout this problem, suppose that the “alphabet” for building a given password is $\{\%, !, ?, A, B, 1, 2\}$.

How many passwords of length 4 can be made if a password must contain at least **one** of each category, and no specific character can be used more than once?

Example: $\{A, A, \%, !\}$ wouldn't be allowed as A is duplicated.

Break this down into three cases

Case i: Symbol Category Repeats

We can think of the forms of this case as **SSLN**. We can rearrange this form using factorial. $\frac{4!}{2!}$. We aren't done yet though, because within a specific form we have a number of options as well!

$$3 S \cdot 2 (L \text{ or } N) \cdot 2 (L \text{ or } N) \cdot 2 S = 24$$

From this we multiply our two results. 12 forms multiplied by 24 arrangements of each of those forms.

$$12 \cdot 24 = 288$$

Case ii: Letter Category Repeats

Here we start the same way as the previous case. We have 12 different forms due to the same logic. As for the arrangements of each form we get:

$$12 \cdot (2L \cdot 3S \cdot 2N \cdot 1L) = 144$$

Case iii: Number Category Repeats

This one is identical to case 2 thankfully, so we know the total passwords we have in this case is also 144.

Adding up all of these passwords together gives us a total of

$$288 + 2(144) = 576$$

How many passwords of length 4 can you create when you have 3 symbols that must be used **at least** once. Say the symbols you have are $\{!, @, \#\}$.

Number of passwords ignoring restrictions: $3^4 = 81$.

In this count are passwords missing an **!**, some missing a **@** and some missing a **#**. We have to be careful though because if we just remove passwords missing, say, **!**, we also risk removing passwords missing other symbols as well. ex: $(@, @, @, @)$ is missing 2 out of the 3 symbols we need.

We will **remove** all passwords that are missing **!**, **@**, **or** **#**.

- A = set of words w/o **!** = 2^4
- B = set of words w/o **@** = 2^4
- C = set of words w/o **?** = 2^4
- E = number of passwords ignoring restrictions = 81

$$|E| - |A \cup B \cup C| = |E| - (|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|)$$

$$|E| - |A \cup B \cup C| = 81 - (2^4 + 2^4 + 2^4 - 1^4 - 1^4 - 1^4 + 0) = \boxed{36}$$

{%, !, ?, A, B, 1, 2}

Length 4 password, at least one from each category.

All possible passwords with no rules: $E = 7^4 = 2401$.

- A = no non-alpha numeric symbol = 4^4
- B = missing a letter = 5^4
- C = missing a number = 5^4

$$|E| - |A \cup B \cup C| = |E| - (|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|)$$

$$|A \cup B \cup C| = 7^4 - (4^4 + 5^4 + 5^4 - 2^4 - 3^4 - 2^4 + 0)$$

Exercise 6d but using set notation!

{%, !, ?, A, B, 1, 2}

No symbol more than once, **and** at least one from each category.

- E = All possible words without repeat: $7 \cdot 6 \cdot 5 \cdot 4 = 840$
- A = no non-alpha numeric symbol = $4!$
- B = missing a letter = $5 \cdot 4 \cdot 3 \cdot 2$
- C = missing a number = $5 \cdot 4 \cdot 3 \cdot 2$
- $A \cap B = 2 \cdot 1 \cdot 0 \cdot 0 = 0$
- $A \cap C = 3 \cdot 2 \cdot 1 \cdot 0 = 0$
- $B \cap C = 2 \cdot 1 \cdot 0 \cdot 0 = 0$
- $A \cap B \cap C = 0^4$

$$840 - (4! + (5 \cdot 4 \cdot 3 \cdot 2) + (5 \cdot 4 \cdot 3 \cdot 2) - 0 - 0 - 0 + 0) = \boxed{576}$$